

Thesis

Why do these apparently unrelated technical experiences belong in the same career?

01. Thesis

I have spent most of my life moving through technical domains that looked unrelated from the outside but kept teaching me the same underlying lesson: complex systems are defined by what flows through them, how those flows are transformed and routed, what controls them, where interfaces change their meaning, and what happens downstream when something changes or fails.

Audio engineering was where that pattern first became explicit to me. A complex audio system is not a collection of independent devices. It is a hierarchy of sources, signal paths, transformations, control signals, interfaces, routing decisions, dependencies, feedback, and consequences. Troubleshooting means understanding the topology well enough to ask the right questions in the right order: Where did the signal originate? What should happen to it next? What transformed it? What controls that transformation? Which path was selected? Where did the observed behavior first diverge from the expected behavior?

That way of thinking became one of the strongest transferable skills I have developed.

Mechanical and automotive work taught me to trace physical causality through engines, drivetrains, electrical faults, and failure modes. Fabrication and welding taught me that a system eventually has to survive material reality: fit, access, sequence, tooling, force, heat, tolerance, repairability, and the difference between what should work on paper and what actually works in the field.

Later technical environments added more layers to the same model. At Level99, I worked around interactive systems involving RFID and other sensors, PLC-monitored elements, DMX, Arduino hardware, programmable LEDs, audio, mechanical components, and computer control. The important lesson was not any one technology. It was the interaction between them. Physical state, sensing, logic, communication, control, and human-visible behavior all had to agree.

Work at the Starlink facility in Bastrop, Texas further sharpened my interest in advanced industrial systems, where integration, sequencing, infrastructure, and reliability operate at a much larger scale. Formal electrical study then gave me another language for the same underlying problem: power, protection, circuit behavior, safety, and controlled physical consequence.

Software and AI extended the pattern again. The medium changed, but the questions did not. What information enters the system? What transforms it? Which layer has authority to act? What context is preserved or lost at an interface? Which routing decision changes the downstream result? How do you distinguish a failure in reasoning from a failure in execution? How do you preserve evidence about what happened well enough to diagnose it afterward?

This is the lens through which I now approach increasingly complex technical systems.

02. Why robotics

Robotics is compelling to me because it compresses nearly every one of those domains into the same machine. A robot is simultaneously a mechanical structure, an electrical system, a collection of sensors and signal paths, a control architecture, a software system, a communications system, a set of interfaces between abstractions, and a physical actor whose outputs change the real world.

That makes robotics less of a departure from my previous work than a convergence point. The through-line is not that I have already mastered every discipline robotics requires - I have not. The through-line is that I have spent years developing the habit of moving between domains, identifying the structure they share, tracing interactions across boundaries, and becoming useful where multiple technical layers meet.

03. How I tend to model systems

Vocabulary

Signal - Transformation - Routing - Control - Interface - Feedback - Consequence

A signal might be audio, voltage, sensor data, a software event, a model output, or a human instruction. A transformation might be amplification, mechanical transmission, computation, interpretation, or policy evaluation. Routing determines where something goes and therefore what can happen next. Control determines which transformations or routes are permitted, selected, or inhibited. Interfaces are where assumptions meet and where meaning is frequently changed, lost, or misinterpreted. Feedback tells the system - or the operator - what actually happened. Consequence is the final reminder that eventually a technical system does something observable, and in robotics that consequence can be physical.

Trajectory: Audio engineering -> Mechanical diagnosis -> Fabrication -> Sensors / controls -> Electrical systems -> Software architecture -> AI orchestration -> Robotics / cyber-physical systems.

04. What I am bringing into robotics

I am not presenting myself as an experienced robotics engineer, embedded-firmware specialist, controls engineer, or someone with years of professional autonomy development. What I can bring immediately is a different combination: hands-on fabrication and welding experience; mechanical and automotive troubleshooting; experience using tools and equipment in real physical environments; electrical fundamentals and formal electrical study; sensor and interactive-system exposure; deep familiarity with signal-path reasoning from audio engineering; software and AI systems thinking; comfort learning across domain boundaries; and a strong bias toward tracing causes rather than treating symptoms.

That combination is still developing. That is part of why I am at RoboBoston. I am looking for environments where I can contribute at the level I have actually earned while continuing to move deeper into robotics, electromechanical systems, integration, test, reliability, controls, and the architecture of systems that have to work outside of a diagram.

05. The general thesis

My work history and technical interests make more sense when treated as one trajectory rather than a collection of unrelated occupations. Audio engineering taught me to see signal paths. Mechanical diagnosis taught me to trace physical causality. Fabrication taught me to respect material reality. Interactive systems taught me to reason across sensors, controls, software, and mechanisms. Electrical study added power, protection, and formal physical-system constraints. Software and AI extended the same reasoning into information, orchestration, and decision systems.

Robotics is where all of those layers become consequential at once. That is the work I am trying to move toward.